

# *Eigenvalues*

## *Eigenvalues and Eigenvectors*

**Defn.** An **eigenvector** of a square matrix  $A$  is a **nonzero** vector  $x$  such that  $Ax = \lambda x$  for some scalar  $\lambda$  called an **eigenvalue**.

## *An Example*

Consider  $A = \begin{bmatrix} 13 & 15 \\ -10 & -12 \end{bmatrix}$ .

Then  $\mathbf{x}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$  is eigenvector for  $\lambda_1 = -2$ , since  $A\mathbf{x}_1 = (2, -2)$ .

And  $\mathbf{x}_2 = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$  is eigenvector for  $\lambda_2 = 3$ , since  $A\mathbf{x}_2 = (-9, 6)$ .

## *Finding Eigenvalues*

Since  $Ax = \lambda x$  is equivalent to  $(A - \lambda I)x = 0$ :

**Fact.** *The value  $\lambda$  is an eigenvalue if and only if the matrix  $A - \lambda I$  is not invertible.*

For example, 0 is an eigenvalue of  $A$  if and only if  $A$  is not invertible.

## *The Characteristic Polynomial*

**Defn.** The **characteristic polynomial** of matrix  $A$  is the determinant of matrix  $A - \lambda I$ . It is a polynomial of degree  $n$ .

**Fact.** The eigenvalues are the roots of the characteristic polynomial.

## *Example Characteristic Polynomial*

Consider again  $A = \begin{bmatrix} 13 & 15 \\ -10 & -12 \end{bmatrix}$ .

This has characteristic polynomial  $\lambda^2 - \lambda - 6$ .

This has roots  $\lambda = 3$  and  $\lambda = -2$ .

## *Eigenspaces*

**Defn.** The **eigenspace** of  $\lambda$  is defined as all eigenvectors corresponding to  $\lambda$  as well as the zero vector.

**Fact.** The eigenspace of eigenvalue  $\lambda$  is the null space of  $A - \lambda I$ .

## *Example Eigenspace*

Consider again  $A = \begin{bmatrix} 13 & 15 \\ -10 & -12 \end{bmatrix}$ . The eigenvalues are  $\lambda_1 = 3$  and  $\lambda_2 = -2$ .

For  $\lambda_1$ , we need a vector in the null space of  $\begin{bmatrix} 15 & 15 \\ -10 & -10 \end{bmatrix}$ ; for example  $(-1, 1)$ .

For  $\lambda_2$ , we need a vector in the null space of  $\begin{bmatrix} 10 & 15 \\ -10 & -15 \end{bmatrix}$ ; for example  $(-3, 2)$ .



## *Another Example*

Consider the matrix

$$\begin{bmatrix} 5 & -4 & -4 \\ -8 & 9 & 8 \\ 10 & -10 & -9 \end{bmatrix}$$

It can be calculated that this has characteristic polynomial  $-\lambda^3 + 5\lambda^2 - 7\lambda + 3$ .

The roots (according to software) are 3, 1, 1.

## *Another Example Continued*

The eigenspaces have bases:

$$\lambda = 1: \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \qquad \lambda = 3: \begin{bmatrix} 2/5 \\ -4/5 \\ 1 \end{bmatrix}$$

## *Eigenvalues of Diagonal Matrices*

**Fact.** *The eigenvalues of a diagonal matrix are its diagonal entries.*

The vectors  $e_i$  are its eigenvectors.

Similarly for a triangular matrix any value on the diagonal is an eigenvalue. However, the eigenvectors need to be calculated by the general method.

## *Summary*

An eigenvector of a matrix  $A$  is a nonzero vector  $x$  such that  $Ax = \lambda x$  where  $\lambda$  is an eigenvalue. The eigenspace of  $\lambda$  is all eigenvectors corresponding to  $\lambda$  as well as the zero vector; it is found by computing the null space of  $A - \lambda I$ .

The characteristic polynomial of matrix  $A$  is the determinant of matrix  $A - \lambda I$ . The eigenvalues are the roots of the characteristic polynomial. The eigenvalues of a diagonal or triangular matrix are its diagonal entries.